Printed Pages: 02 **Sub Code: RPH-302**

Paper Id: 233502 **Roll No.** | | | | | | | | | | |

B. PHARM.
(SEM III) THEORY EXAMINATION 2022-23
PHARMACEUTICS-II (UNIT OPERATIONS)

Time: 3 Hours ***Total Marks: 70***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 7 = 14

a. Distinguish between Primary Units & Secondary Units.
b. Compare Raw Water and Purified Water.
c. Filter aids increases the rate of filtration. How?
d. Define Bound Moisture and Unbound Moisture.
e. Illustrate under-driven & over-driven assembly?
f. Differentiate between primary refrigerant and secondary refrigerant.
g. Give two methods of prevention of corrosion.

SECTION B

2. Attempt any *three* of the following: 7 x 3 = 21

a. Discuss about the advantages of automated process control system. Add a note on Process Variables.
b. Compare & Contrast theories of Filtration. Add a note on applications of filtration.
c. Categorize the types of Dryers? Describe the construction, working and uses of Fluidized Bed Dryer.
d. How will you determine humidity & relative humidity of air by using psychrometer and psychrometric chart? Explain.
e. Summarize the general composition of Stainless Steel? Classify the types of stainless steel used in industries.

SECTION C

3. Attempt any *one* part of the following: 7 x 1 = 7

(a) Explain the terms Mass Balance and Energy Balance with suitable examples.
(b) Define Process Variable. Write a descriptive note on Temperature Measurement.

4. Attempt any *one* part of the following: 7 x 1 = 7

(a) Describe the principle, construction, working and applications of Perforated Basket Centrifuge.
(b) Write a descriptive note on quality requirement or treatment of water.

5. Attempt any *one* part of the following: 7 x 1 = 7

(a) Write a descriptive note on Freeze Drying. Discuss its pharmaceutical

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applications also.

(b) Describe the principle, construction, working and applications of Spray Drying.

6. Attempt any *one* part of the following: 7 x 1 = 7

(a) Discuss about the approaches for humidification & dehumidification with relevant labeled diagram.

(b) Define the following terms:
i) Relative Humidity ii) DBT iii) Dew Point iv) WBT

7. Attempt any *one* part of the following: 7 x 1 = 7

(a) Classify material of construction. Write a descriptive note on Gasses used in pharmaceutical industry.

(b) Explain the types of hazards occur due to electricity. Discuss about its preventive measures.

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